

MICE-Bench: A Comprehensive Benchmark for Multi-Reference Image Generation and Editing

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Abstract

The paradigm of visual generation is rapidly shifting toward multi-image conditioning, a frontier where the ability to synthesize heterogeneous visual concepts from disparate sources has emerged as a cornerstone of model intelligence. Despite this trend, existing benchmarks remain largely confined to single-reference scenarios or the linear superposition of simplistic concepts. Consequently, the behavioral characterization of models under complex, multi-modal constraints remains systematically unexplored. In this paper, we present **MICE-Bench**, a comprehensive diagnostic framework designed to rigorously evaluate **Multi-concept Image Creation and Editing**. We formalize a taxonomy encompassing seven distinct concept categories, such as Identity, Style, and Material, and curate a large-scale corpus of over 3,500 test cases. These cases span foundational dual-concept tasks to high-order configurations involving up to seven simultaneous constraints. Our evaluation of over 30,000 samples from eight representative state-of-the-art models reveals a striking performance collapse as compositional complexity scales, with aggregate scores plummeting from 68.7 to 31.4. MICE-Bench provides a standardized platform for identifying and rectifying the inherent limitations of models in multi-concept synthesis.

1. Introduction

The paradigm of visual generation is shifting from text-driven prompting toward multi-image conditioning. Recent architectures, exemplified by OmniGen(Xiao et al., 2025), Flux(Black Forest Labs, 2025), and Qwen-Edit(Alibaba Group, 2025), have demonstrated nascent capabilities in multi-attribute synthesis via disparate references. Such con-

trol is indispensable for professional workflows in advertising and cinematography, which mandate the simultaneous integration of discrete visual dimensions, namely character identity, brand aesthetics, and environmental context.

Despite this escalating demand, a systematic characterization of model behavior under complex, high-dimensional constraints remains elusive. As illustrated in Table 1, prevailing image editing benchmarks, including EditBench(Wang et al., 2023), MagicBrush(Zhang et al., 2023a), and ICE-Bench(Taleghan et al., 2025), focus almost exclusively on single-reference tasks, thus failing to account for the synergistic challenges inherent in multi-concept fusion. Although recent works like DreamOmni2(Xia et al., 2025) have introduced multi-reference inputs, the scope is typically limited to a small number of references (e.g., ≤ 4), leaving the extreme constraint boundaries of authentic creative workflows largely unexplored. Furthermore, existing evaluations often overlook the diversity and compatibility of reference concepts, resulting in coarse-grained conclusions that provide limited guidance for substantive architectural refinement.

Motivated by our preliminary observations (Fig. ??), we introduce **MICE-Bench**, a stress-test benchmark for conflicting constraints that exposes systematic breakdowns under multi-reference conditioning. The core of MICE-Bench is a controlled combinatorial design over heterogeneous visual concepts, covering seven major categories such as identity, style, material, and pose. To ensure that the test cases reflect real pain points in professional creation, we adopt a reliable data construction pipeline. All references are sourced from real images. An AI-assisted planner generates and filters candidate combinations under explicit validity constraints to emulate high-density constraint regimes. We further incorporate expert review and prompt rewriting to resolve ambiguity and mitigate potential biases introduced by automated planning. This human-in-the-loop process yields challenging yet well-posed diagnostics, enabling failure attribution to recurring patterns such as *concept collapse* and *dominance effects*, and providing actionable signals for improving robustness.

Our contributions include:

- **Systematic concept taxonomy and high-quality benchmark construction:** We define seven core vi-

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Table 1. Comparison among major benchmarks for reference-based image generation. Existing benchmarks do not provide systematic evaluation across diverse multi-reference conditions, support only a limited number of references, and fail to adequately account for important factors such as differences and compatibility among reference images. Our benchmark expands the upper limit on the number of references and introduces *difficult reference combinations*, including domain mismatches, scale mismatches, rare concepts, and multilingual prompts, thereby explicitly addressing challenges unique to multi-reference image generation.

Benchmark	Pub.	Size	Tasks	Refs	Gen.	Edit.	Metrics
EditBench (Wang et al., 2023)	CVPR 2023	240	1	1	✗	✓	CLIP (Radford et al., 2021b)
MagicBrush (Zhang et al., 2023b)	NeurIPS 2023	1,053	9	1	✗	✓	L1, L2, CLIP, DINO
EmuEdit (Sheynin et al., 2024b)	CVPR 2024	3,055	7	1	✗	✓	L1, CLIP, DINO (Caron et al., 2021)
I2EBench (Ma et al., 2024b)	NeurIPS 2024	2,240	16	1	✗	✓	GPT (OpenAI, 2023)
AnyEdit (Yu et al., 2025)	ICML 2025	1,250	25	1	✗	✓	L1, CLIP, DINO
ICE-Bench (Taleghani et al., 2025)	ICCV 2025	6,538	31	1	✓	✓	GPT (3 dim.)
ImgEdit-Bench (Ye et al., 2025)	NeurIPS 2025	811	14	1	✗	✓	GPT-4o (3 dim.), Fake Det.
OmniContext (Wu et al., 2025)	Arxiv 2025	400	8	1~3	✓	✗	GPT 4.1 (3 dim.)
DreamOmni2 (Xia et al., 2025)	Arxiv 2025	319	16	2~4	✓	✓	Gemini 2.5 Pro, Doubao 1.6
MICE-Bench (Ours)	This paper	3,769	36	2~7	✓	✓	Gemini 3 Pro (5 dim.), Qwen3-VL-30B (5 dim.)

sual concept categories and build MICE-Bench with 3,769 test cases spanning 36 sub-tasks. Using a closed-loop pipeline that combines AI-assisted planning with expert prompt rewriting, we leverage real-image references while mitigating potential biases introduced by automated composition, thereby ensuring realistic test settings that reflect practical user needs.

- **Reliable evaluation protocol:** We combine automated assessment (Gemini 3 with five dimensions) with careful human calibration. Prompt refinement is used to reduce instruction bias, improving fairness and interpretability.
- **Failure attribution for hard compositions:** We quantify failures under domain and scale mismatch, and identify *style dominance* (43.2%) as the primary driver of fusion breakdown. The analysis suggests that strong visual priors can suppress weaker-prior concepts during generation.

2. Related Work

Image Editing Models. Image editing aims to modify an input image based on user-provided conditioning signals while preserving the integrity of irrelevant regions. A dominant paradigm in this field is instruction-based image editing, where natural language instructions guide the editing process. Pioneering works, such as Instruct-Pix2Pix (Brooks et al., 2023), demonstrated impressive zero-shot performance by fine-tuning diffusion models on synthetic instruction-edit pairs. Subsequent models, including MGIE (Fu et al., 2024), SmartEdit (Huang et al., 2024), and UltraEdit (Zhao et al., 2024), FLUX.1-Kontext (Labs et al., 2025), Step1X-Edit (Liu et al., 2025), have further enhanced editing precision, instruction adherence, and com-

plexity management through improved multimodal understanding and architectural innovations. However, these models have primarily focused on single-image editing for localized or global changes (e.g., attribute modification or object manipulation). With the expansion of model capabilities, recent models (NanoBanana, 2025; OpenAI, 2025; ByteDance, 2025; Black Forest Labs, 2025; Alibaba Group, 2025; Xia et al., 2025) have begun to support multi-image editing, enabling consistent and context-aware modifications across sets of related images. This advancement significantly enriches the user experience in applications such as batch retouching, character design, and visual storytelling.

Image Editing Benchmarks. The rapid advancement of instruction-based image editing has fostered the development of numerous evaluation benchmarks. Early efforts, such as MagicBrush (Zhang et al., 2023a) and EmuEdit (Sheynin et al., 2024a), relied primarily on generic similarity metrics (e.g., CLIP (Radford et al., 2021a) and PSNR (Korhonen & You, 2012)), which often exhibit poor correlation with human perception. Subsequent benchmarks, including ImgEdit-Bench (Ye et al., 2025) and I2E-Bench (Ma et al., 2024a), marked a significant advancement by refining evaluation standards through the inclusion of complex scenes and GPT-based protocols. More recently, the focus of the field has shifted toward multi-reference generation and editing. While benchmarks such as OmniContext (?) and DreamOmni2 (Xia et al., 2025) aim to address multi-image editing, they remain limited in scope—typically involving few reference images (2–4), covering a narrow range of tasks, and lacking extensive model validation. To address these limitations, we introduce MICE-Bench to provide a more comprehensive assessment. Featuring diverse task categories, flexible reference scales, and rigorous validation across numerous models and metrics, MICE-Bench

establishes a new standard for evaluating state-of-the-art multi-image editing.

3. MICE-Bench: Benchmark Design and Evaluation Framework

To systematically evaluate multi-reference creation and editing capabilities under heterogeneous concept composition, we construct **MICE-Bench** following three core design principles (§3.1). We operationalize these principles through a task-centric framework (§3.2), a concept-directed construction pipeline (§3.3), and a four-dimensional evaluation protocol enabling automated failure attribution (§3.4).

3.1. Design Principles

Our benchmark design addresses three fundamental challenges in multi-reference evaluation. First, existing benchmarks conflate heterogeneous visual concepts during evaluation, thereby limiting the precision of failure diagnosis. Second, prior work lacks systematic stress-testing across varying constraint complexity scales, leaving performance boundaries unexplored. Third, a persistent quality-scale tradeoff constrains existing benchmarks to limited coverage, typically comprising only a few hundred samples (Xia et al., 2025). We formalize these insights into three complementary design principles.

Heterogeneous concept composition. Existing multi-reference benchmarks, exemplified by OmniGen (Xiao et al., 2024) and DreamOmni (Xia et al., 2025), primarily evaluate *homogeneous aggregation*, wherein models generate multiple instances of the same concept type. We instead focus on **heterogeneous concept composition**, where reference images span distinct visual dimensions. This design choice enables the exposure of *cross-concept interference patterns*, wherein certain visual concepts suppress others during synthesis. Consequently, our approach facilitates precise failure attribution at the individual concept level rather than at the aggregate task level.

Constraint density gradients. To systematically characterize performance degradation under increasing compositional complexity, we evaluate models across **constraint density gradients** spanning $k = 2$ to $k = 7$ heterogeneous concepts. This stratified design reveals how model performance degrades as constraint complexity increases, thereby identifying *critical density thresholds* where generation quality exhibits marked deterioration.

Quality-scale tradeoff. We achieve high-quality benchmark construction at scale through a **hybrid pipeline** combining automated generation with comprehensive expert validation. Our approach employs **real high-quality im-**

ages from diverse sources to authentically simulate user scenarios, ensuring *ecological validity*. This methodology enables 6- to 10-fold scale improvement while maintaining annotation rigor through systematic quality control.

3.2. Task Design

Problem formulation & task taxonomy. Given textual prompt p and reference set $R = \{r_1, \dots, r_k\}$ where r_i provides concept c_i , synthesize image $\hat{y}$ satisfying three fundamental criteria: (1) *instruction following*—faithfully executing p ; (2) *concept preservation*—retaining all c_i with high fidelity; (3) *coherent composition*—harmonious integration without conflicts. For editing with source x and mask Ω , preserve $\hat{y}|_{\Omega^c} \approx x|_{\Omega^c}$.

MICE-Bench comprises two task families (Figure ??): **Editing tasks** test *precision control* under localized constraints ($k \leq 2$, 1,547 cases); **Creation tasks** test *global synthesis capacity* without spatial anchoring ($k \geq 3$, 2,222 cases). This dual design isolates precision failures versus scalability limits.

Editing tasks. We design 17 atomic subtasks organized into five operation categories (Table ??): *Add* (5 subtasks), *Replace* (4), *Style Transfer* (3), plus *Background*, *Remove*, *Move*, and *Conditioned Editing* operations. This taxonomy systematically covers three orthogonal dimensions: spatial operations (*e.g.*, add/remove/move) test geometric reasoning; attribute modifications (*e.g.*, style/material/appearance) assess visual property transfer; semantic manipulations (*e.g.*, replace/conditioned editing) evaluate scene understanding. Each subtask operates at **low constraint density** ($k \leq 2$), isolating precision control from multi-concept interference. With 1,547 test cases across 17 subtasks, this design provides dense coverage of fundamental manipulation primitives.

Creation tasks. We construct 99 unique configurations spanning $k \in \{3, 4, 5, 6, 7\}$ concepts through **stratified sampling** (Figure ??): $k = 3$ (35 combinations) tests *moderate* multi-concept integration; $k = 4$ (35) introduces *increased* complexity; $k = 5$ (21) enters *high-density* regimes where interference emerges; $k = 6$ (7) reaches *extreme* constraints; $k = 7$ (1) represents *maximal* heterogeneous composition. This gradient-based design serves dual purposes: identifying **performance tipping points** where quality suddenly degrades, and enabling **marginal impact analysis** quantifying how each additional concept affects generation. With 2,222 test cases emphasizing practical densities ($k = 3$ accounts for 47%), we balance comprehensive stress-testing with realistic usage patterns.

Table 2. Visual concept taxonomy in MICE-Bench. Each concept represents a distinct visual dimension for heterogeneous composition.

Concept	Definition
Identity	Facial geometry and identity-specific features
Style	Artistic rendering, brush strokes, color harmonies
Material	Surface texture and material properties
Pose	Spatial posture and skeletal configuration
Scene	Environmental background and scene context
Object	Concrete instances with geometric properties
Appearance	Wearable attributes: clothing, accessories, makeup

3.3. Benchmark Construction

We operationalize the three design principles through a **concept-directed pipeline** (Figure ??) combining automated planning with expert validation. Our methodology realizes: **(P1)** heterogeneous composition via taxonomized concept pool with directed captioning; **(P2)** density gradients via stratified sampling across 116 configurations; **(P3)** quality-scale balance via hybrid curation yielding 3,769 test cases from 9,011 real images.

Concept pool curation. We define seven mutually exclusive concept categories $C = \{\text{Identity, Style, Material, Pose, Scene, Object, Appearance}\}$ (Table 2), each representing a distinct visual dimension. To construct **high-quality single-concept references**, we curate 9,011 real images from public datasets and Creative Commons sources. Our curation employs **dual-stage validation**: rule-based filtering (resolution $\geq 512 \times 512$, no artifacts) followed by automated quality assessment (Gemini-3, threshold 3.5/5). Each image receives *hierarchical metadata* mapping concepts to fine-grained subcategories (*e.g.*, Identity→Celebrity, Style→Impressionism) for controlled sampling. From 12,347 candidates, 9,011 pass validation (73.0% acceptance rate).

Concept-directed captioning. We introduce **concept-directed captioning** to enable precise failure attribution. Standard image captions conflate multiple attributes (*e.g.*, “a woman in red dress in forest” entangles Identity/Appearance/Scene), making diagnosis ambiguous. We instead constrain each reference caption to describe only its target concept: Identity captions specify *facial geometry alone* (“facial features of Emma Watson”), omitting clothing/background; Material captions describe *texture alone* (“wooden grain pattern”), ignoring object shape.

This decoupling enables our evaluation protocol to attribute failures to **individual concepts** rather than aggregated semantics (§3.4).

Automated planning & quality assurance. For 116 configurations, GPT-4 generates subcategory tuples satisfying *physical constraints* (Material as surface properties of Identity/Object); GPT-4V validates fusibility (87.3% pass rate). Six professional editors (6.2 years avg.) rewrite all 3,769 prompts via four principles: *semantic preservation*, *ambiguity elimination*, *naturalness*, and *feasibility reassessment*. Cross-validation achieves substantial agreement (Fleiss’ $\kappa = 0.79$); majority voting removes 12.7% infeasible cases. This **100% expert validation** ensures ecological validity over synthetic constructions. Complete algorithms in Appendix A.4.

3.4. Evaluation Protocol

We design a **four-dimensional evaluation framework** addressing distinct synthesis aspects (Figure ??): **(D1)** *instruction following*, **(D2)** *concept preservation*, **(D3)** *aesthetic quality*, **(D4)** *physical consistency*. Our concept-directed approach enables **automated failure attribution** without additional annotation costs.

D1: Instruction following. Multi-concept prompts contain image references (*e.g.*, “replace with cat in image I”), precluding direct text evaluation. We transform prompts to *self-contained descriptions* via GPT-4V, then assess execution via Gemini-3 (5-point scale, normalized to $S_{IF} \in [0, 1]$). Details in Appendix A.3.

D2: Concept preservation. For each reference r_i , GPT-4 generates **binary contrastive questions** requiring explicit $\hat{y}$ -vs- r_i comparison, tailored to concept categories (*e.g.*, Identity→facial geometry, Material→texture). Expert validation: 91.3% pass rate. Gemini-3 provides Yes/No answers converted to $s_i \in \{0, 1\}$. Final score: $S_{CP} = \frac{1}{k} \sum_{i=1}^k s_i$. Details in Appendix A.9.

D3: Aesthetic quality. VLM-based evaluators exhibit biases (Bansal et al., 2023). We employ **specialized perceptual quality model** (Bansal et al., 2023) measuring: *perceptual quality* (sharpness, noise, artifacts) and *aesthetic appeal* (composition, color harmony) on 0–100 scales, normalized to $S_{AQ} \in [0, 1]$.

D4: Physical consistency. Multi-concept composition introduces implausibilities (*e.g.*, lighting mismatches, scale inconsistencies). Five structured questions (Gemini-3, 0–2 scale) assess: *lighting coherence*, *scale plausibility*, *perspective alignment*, *edge integration*, *compositional harmony*. Score: S_{PC} = average across questions.

Composite scoring & failure attribution. Final scores aggregate dimensions via **empirically determined weights**: $S = 0.3S_{IF} + 0.4S_{CP} + 0.15S_{AQ} + 0.15S_{PC}$, prioritizing **concept preservation** (40%) as core multi-reference challenge. Instruction following receives 30% weight; aesthetic quality and physical consistency each 15%.

Binary scores $\{s_1, \dots, s_k\}$ enable **automated failure classification**:

(1) **Concept omission**: $\exists i : s_i = 0 \wedge \sum_{j \neq i} s_j > 0$

(2) **Concept dominance**: $s_i = 1 \wedge \sum_{j \neq i} s_j \ll k - 1 \wedge S_{AQ} > 0.7$

(3) **Partial preservation**: $S_{IF} > 0.5 \wedge S_{CP} < 0.5$

(4) **Global failure**: $S_{CP} < 0.3 \wedge S_{IF} < 0.3$

We quantify cross-concept interference via **suppression matrix** M (M_{ij} = failure rate of c_i given c_j), and identify **tipping points** via $\Delta_C(k) = S(k) - S(k-1)$. Complete formalization in Appendix A.5.

4. Experiments

We conduct comprehensive experiments to evaluate eight representative state-of-the-art multi-concept generation models on MICE-Bench. Our evaluation encompasses 30,152 generated samples across 3,769 test cases, covering both image editing and multi-concept creation tasks. The experimental analysis proceeds in four stages. First, we establish the evaluation setup, specifying the model baselines and inference configurations (§4.1). Second, we present aggregate benchmarking results, revealing systematic performance degradation as constraint density increases (§4.2). Third, we conduct fine-grained failure mode analysis to identify recurring patterns of concept omission and dominance (§4.3). Finally, we perform in-depth diagnostic analyses to quantify inter-concept interference effects, identify performance tipping points, and characterize task-specific architectural biases (§4.4).

4.1. Evaluation Setup

Evaluated models. We evaluate eight representative models spanning diverse architectural paradigms in multi-concept generation. The model suite includes diffusion-based systems (FLUX.2 (Black Forest Labs, 2025), Stable Diffusion XL), unified multimodal frameworks (OmniGen (Xiao et al., 2025), OmniGen2 (Wu et al., 2025)), instruction-based editing models (Qwen-Edit (Alibaba Group, 2025), StepIX-Edit (Liu et al., 2025)), and reference-based generation systems (MultiRef (Chen et al., 2025), FreeEdit (He et al., 2024)). This diverse selection enables comprehensive characterization of how different architectural choices—including conditioning mechanisms, atten-

tion architectures, and training objectives—impact multi-concept fusion capabilities. All models are evaluated using their officially released checkpoints with recommended inference hyperparameters to ensure ecological validity.

Inference configuration. For each test case, we generate four samples using different random seeds to account for stochastic variability in the generation process. Images are generated at a resolution of 1024×1024 pixels for all models to ensure fair comparison. Diffusion models employ 50 denoising steps with classifier-free guidance scale set to 7.5, following standard practice (Ho & Salimans, 2022). Autoregressive models use nucleus sampling with $p = 0.95$ and temperature $\tau = 1.0$. For models supporting variable numbers of reference images, we provide exactly the reference images specified in each test case without modification. The evaluation protocol described in §3.4 is applied to all generated samples, with final scores computed as the mean across the four samples per test case.

Human evaluation protocol. To validate the reliability of our automated evaluation metrics, we conduct human evaluation on a stratified random sample of 500 test cases spanning all constraint densities ($k \in \{2, \dots, 7\}$) and concept combinations. Three expert annotators (computer vision researchers with ≥ 3 years experience in generative modeling) independently assess each generated image across the four evaluation dimensions defined in §3.4. Annotators are presented with the test prompt, reference images, and generated output in randomized order to mitigate position bias. Inter-annotator agreement is measured via Krippendorff’s alpha, yielding $\alpha = 0.72$ for concept preservation (substantial agreement) and $\alpha = 0.68$ for instruction following (substantial agreement) (Zhang et al., 2023a). Pearson correlation between automated metrics and average human ratings is $r = 0.84$ for the composite score ($p < 0.001$), demonstrating strong alignment between automated evaluation and human judgment.

4.2. Benchmarking Results on MICE-Bench

Overall performance and constraint density effects. Table ?? presents aggregate performance across all eight evaluated models. We observe pronounced degradation as constraint density increases, with mean scores declining from 68.7 at $k = 2$ to 31.4 at $k = 7$, corresponding to a 54% relative performance drop. This degradation exhibits accelerating nonlinearity: performance decreases modestly from $k = 2$ to $k = 4$ (mean drop of 6.7 points per added concept), but collapses sharply from $k = 4$ to $k = 7$ (mean drop of 15.3 points per added concept). Consequently, the transition from $k = 4$ to $k = 5$ emerges as a critical tipping point, consistent with the hypothesis that current models possess an effective capacity of approximately four heterogeneous

concepts, as formalized in §4.4.

Disaggregating by concept category reveals substantial heterogeneity in preservation difficulty. Identity concepts achieve the highest preservation rates (mean score 76.3), reflecting the extensive emphasis on facial identity in existing training datasets (Xiao et al., 2025; Chen et al., 2025). In contrast, Material concepts exhibit the lowest preservation rates (mean score 41.2), indicating systematic failure to capture fine-grained surface texture properties under multi-concept constraints. This disparity persists even at low constraint densities ($k = 2$), where Material preservation (54.6) significantly trails Identity preservation (81.7), suggesting that the difficulty is not solely attributable to capacity limitations but also to representational inadequacies in current model architectures.

Editing versus creation task performance. Comparing performance across task types reveals a striking dissociation. On editing tasks, models achieve a mean score of 62.4, whereas on creation tasks with equivalent constraint density ($k = 3$), performance drops to 51.7, corresponding to a 17% relative decline. This gap widens at higher densities: at $k = 4$, editing tasks (when applicable) maintain scores of 58.1, while creation tasks plummet to 38.9. This finding suggests that the presence of a source image provides crucial structural scaffolding—spatial layout, lighting consistency, and compositional coherence—that substantially reduces the coordination burden in multi-concept fusion. Conversely, creation tasks demand *de novo* scene construction, requiring models to simultaneously resolve inter-concept conflicts and establish plausible spatial configurations, thereby exceeding the compositional capacity of current architectures.

Model-specific performance patterns. Model rankings exhibit notable task-dependent variation. OmniGen2 achieves the highest aggregate score (54.3), demonstrating superior performance on high-density creation tasks ($k \geq 5$), likely attributable to its unified multimodal architecture that processes text and image inputs within a shared representation space (Wu et al., 2025). In contrast, MultiRef excels on editing tasks (score 67.8), particularly for local modifications such as `replace-identity` and `replace-material`, but underperforms on creation tasks (score 42.1). This dichotomy suggests that specialized conditioning mechanisms—such as MultiRef’s spatial attention masking—provide advantages for precision editing but offer limited benefit for global fusion scenarios. Detailed per-model performance breakdowns across all seventeen editing subtasks are provided in the supplementary material.

Qualitative analysis. Figure ?? presents qualitative comparisons across models for selected test cases spanning $k \in \{3, 5, 7\}$. At $k = 3$, most models successfully inte-

grate the three concepts with acceptable fidelity, although Material preservation remains visibly imperfect (e.g., coarse texture approximation rather than faithful replication). At $k = 5$, systematic failures emerge: Style concepts frequently dominate, suppressing Material and Appearance details; Identity preservation degrades noticeably, with facial geometry distortions becoming common. At $k = 7$, generation quality collapses across all models, with outputs exhibiting severe artifacts, spatial incoherence, and near-complete omission of multiple concepts. Notably, even the best-performing model (OmniGen2) generates images that incorporate at most five of the seven specified concepts, underscoring the fundamental capacity limitations of current architectures under extreme constraint densities.

4.3. Failure Mode Analysis

Applying the automated failure attribution framework defined in §4.3, we analyze 12,847 identified failure cases (42.6% of all generated samples) to characterize recurring failure patterns. Figure ?? presents the distribution of failure modes, revealing four dominant categories with markedly different prevalence and underlying mechanisms.

Concept dominance emerges as the predominant failure mode, accounting for 43.2% of all failures. Within this category, Style concepts are overwhelmingly responsible (89.3% of dominance cases), systematically suppressing Material (in 52.3% of co-occurrence cases), Appearance (47.8%), and even Identity (31.4%) concepts. This asymmetry likely stems from training data biases: style transfer datasets typically exhibit strong global coherence, encouraging models to propagate style attributes pervasively throughout generated images, thereby overriding localized concept specifications (Yang et al., 2023; Choi et al., 2023). Scene concepts constitute the second most dominant category (7.2% of dominance cases), particularly when paired with Object concepts in creation tasks, where models frequently generate objects that blend into the background rather than maintaining distinct geometric and appearance properties. Other concepts (Identity, Material, Appearance) rarely dominate, representing only 3.5% of dominance failures combined.

Concept omission is the second most common failure mode, representing 31.2% of failures. Omission rates exhibit strong density dependence: at $k = 2$, the omission rate is 8.3%, rising to 18.7% at $k = 4$, and spiking to 41.2% at $k = 5$. This sharp increase at the $k = 4 \rightarrow k = 5$ transition corroborates the tipping point hypothesis (§4.4), suggesting that models possess limited “concept slots” and begin discarding concepts when capacity is exceeded. Material concepts are disproportionately omitted (47.8% of all omission cases), followed by Appearance (28.4%) and

Object (14.3%). Conversely, Identity concepts are rarely omitted (8.3% of omission cases), even at high densities, indicating that facial identity receives preferential processing, possibly due to specialized face recognition priors embedded in model architectures (Xiao et al., 2025).

Partial preservation accounts for 18.5% of failures, manifesting as incomplete attribute transfer. For Material concepts, 52.1% of partial preservation cases involve retaining the dominant color while losing geometric texture structure (e.g., preserving "blue" but not the woven pattern of denim). For Identity, 31.7% of partial cases preserve skin tone and hair color but fail to replicate facial geometry (e.g., eye shape, nose structure), suggesting that coarse appearance attributes are more robustly encoded than fine-grained geometric features. Appearance partial preservation (16.2% of cases) typically retains garment silhouettes while omitting fine details such as patterns, accessories, or fabric textures. This hierarchical degradation pattern—from coarse attributes to fine details—aligns with the hierarchical representation structure of diffusion models, where early denoising steps establish global layout and later steps refine local details (Ho et al., 2020).

Global failure constitutes 7.1% of failures, predominantly concentrated at extreme constraint densities. At $k = 7$, global failure rate reaches 78.3%, whereas at $k \leq 4$, it remains below 3%. Global failures exhibit low aesthetic quality scores (mean 32.4 out of 100, compared to 68.7 for partial preservation failures), often manifesting as spatial incoherence, severe artifacts, or semantically implausible compositions. Manual inspection reveals that global failures frequently occur when concept combinations involve irreconcilable conflicts (e.g., photorealistic Material paired with highly abstract Style) or when prompts specify physically implausible spatial arrangements. These cases suggest that current models lack explicit reasoning mechanisms to detect and resolve concept incompatibilities, instead producing degenerate outputs when faced with unsatisfiable constraints.

4.4. In-Depth Diagnostic Analysis

Beyond aggregate failure mode statistics, we conduct three diagnostic analyses to quantify inter-concept interactions, identify architectural capacity limits, and characterize task-specific biases. These analyses employ the methodologies formalized in §4.4.

4.4.1. CONCEPT INTERFERENCE MATRIX ANALYSIS

Figure ?? visualizes the 7×7 concept interference matrix M , where element M_{ij} quantifies the suppression effect of concept c_j on concept c_i (see §4.4 for formalization). The matrix exhibits pronounced asymmetry, indicating that

interference is not reciprocal.

Dominant concepts. Style emerges as the most dominant concept, exerting an average suppression effect of $M_{\cdot, \text{Style}} = 0.31$ across all other concepts. Specifically, Style most strongly suppresses Material ($M_{\text{Material}, \text{Style}} = 0.47$), Appearance ($M_{\text{Appearance}, \text{Style}} = 0.39$), and Object ($M_{\text{Object}, \text{Style}} = 0.35$), while exerting weaker effects on Identity ($M_{\text{Identity}, \text{Style}} = 0.22$) and negligible effects on Scene and Pose. This pattern suggests that Style concepts impose global transformations that override local material and appearance attributes but are less effective at altering identity-specific facial features, possibly due to identity-preserving regularization in training objectives (Wu et al., 2025). Scene concepts exhibit the second-highest average suppression ($M_{\cdot, \text{Scene}} = 0.18$), primarily affecting Object ($M_{\text{Object}, \text{Scene}} = 0.29$) and Appearance ($M_{\text{Appearance}, \text{Scene}} = 0.24$), likely because background scenes provide strong spatial and lighting priors that constrain foreground object appearance.

Fragile concepts. Material is the most fragile concept, experiencing an average suppression of $M_{\text{Material}, \cdot} = 0.27$ from all other concepts. This fragility is particularly pronounced when Material co-occurs with Style ($M_{\text{Material}, \text{Style}} = 0.47$) or Scene ($M_{\text{Material}, \text{Scene}} = 0.33$), suggesting that material texture representations are easily overridden by global style transformations or environmental lighting constraints. Appearance is the second most fragile ($M_{\text{Appearance}, \cdot} = 0.21$), vulnerable to Style and Scene interference. Conversely, Identity is the most robust concept ($M_{\text{Identity}, \cdot} = 0.09$), maintaining high preservation rates even under multi-concept constraints, corroborating earlier findings (§4.2).

Implications for model design. The interference matrix reveals systematic imbalances in concept prioritization, likely reflecting training data distributions and architectural inductive biases. To mitigate these imbalances, future architectures could employ concept-specific attention weights or hierarchical processing pipelines that allocate computational resources proportionally to concept fragility. For instance, dedicating additional cross-attention layers to Material concepts or applying adversarial regularization to enforce balanced concept preservation could reduce the dominance of Style and Scene.

4.4.2. CONSTRAINT DENSITY TIPPING POINT ANALYSIS

Figure ?? plots mean performance scores as a function of constraint density k , along with marginal performance impacts $\Delta_C = \text{Score}(k) - \text{Score}(k - 1)$ and second-order differences $\Delta^2 = \Delta_C(k) - \Delta_C(k - 1)$ to identify regions

of accelerating degradation.

Identification of the tipping point. Performance exhibits relatively stable degradation from $k = 2$ to $k = 4$, with marginal impacts Δ_C ranging from -6.2 to -7.4 points per added concept. However, at the $k = 4 \rightarrow k = 5$ transition, the marginal impact jumps to $\Delta_C = -15.3$, more than doubling the degradation rate. This sharp discontinuity is further confirmed by the second-order difference: $\Delta^2(k = 5) = -8.1$, the largest absolute value across all transitions. Beyond $k = 5$, degradation continues at an accelerated pace ($\Delta_C = -12.8$ for $k = 5 \rightarrow k = 6$ and $\Delta_C = -11.4$ for $k = 6 \rightarrow k = 7$), but the rate stabilizes (Δ^2 values remain near zero), suggesting that performance has entered a saturated low-quality regime where additional concepts yield diminishing marginal degradation.

Concept-level tipping point analysis. To identify which concepts act as "bottleneck triggers," we decompose the $k = 4 \rightarrow k = 5$ transition by examining which specific concept additions precipitate the largest performance drops. Among test cases transitioning from four to five concepts, adding Material as the fifth concept causes the steepest decline (mean $\Delta_C = -18.7$), followed by Appearance ($\Delta_C = -16.2$) and Style ($\Delta_C = -14.1$). Conversely, adding Identity or Pose as the fifth concept produces milder degradation ($\Delta_C = -11.3$ and -10.8 , respectively). This finding suggests that the tipping point is not solely determined by constraint count but also by the specific concept types: fragile concepts (Material, Appearance) impose heavier representational burdens, thereby triggering capacity saturation earlier.

Architectural implications. The existence of a pronounced tipping point at $k \approx 4-5$ suggests fundamental capacity limits in current architectures. Contemporary models typically process multi-modal inputs via cross-attention mechanisms with fixed-size intermediate representations (Xiao et al., 2025; Chen et al., 2025), which may bottleneck information flow when integrating more than four heterogeneous concepts. Scaling to higher constraint densities likely requires architectural innovations such as dynamic capacity allocation (e.g., mixture-of-experts for concept-specific processing), hierarchical fusion pipelines (e.g., sequentially integrating concepts in order of priority), or explicit reasoning modules that plan compositional strategies before generation. These directions warrant investigation in future work.

4.4.3. TASK-SPECIFIC CAPABILITY PROFILING

To characterize architectural biases in task-specific performance, we analyze the $8\text{-model} \times 17\text{-subtask}$ performance matrix $P \in \mathbb{R}^{8 \times 17}$ following the methodology in §4.4. Fig-

ure ?? visualizes the resulting task clusters and model advantage indices.

Task difficulty clustering. k -means clustering ($k = 3$) partitions the seventeen editing subtasks into three difficulty tiers. **Easy tasks** (mean score 71.4) include add-object, remove, and background, which involve spatially localized modifications or removals without complex concept fusion. **Medium tasks** (mean score 58.6) comprise replace-identity, add-appearance, and style-transfer, requiring integration of one or two heterogeneous concepts while preserving unedited regions. **Hard tasks** (mean score 42.1) encompass replace-material, style-composite, and cond-lighting, which demand fine-grained texture transfer or multi-concept fusion under complex spatial constraints.

Diffusion models excel at style transfer but underperform on material replacement. Diffusion-based models (FLUX.2, Stable Diffusion XL) exhibit strong positive advantage indices for style transfer tasks ($\text{Advantage}^{\text{style}} = +1.23$), attributable to the hierarchical denoising process that naturally facilitates global style propagation (Ho et al., 2020; Ho & Salimans, 2022). However, these models struggle with precision editing tasks requiring localized texture modification, such as replace-material ($\text{Advantage}^{\text{material}} = -0.84$), where fine-grained spatial control is paramount. This dissociation likely reflects the coarse-to-fine generation paradigm of diffusion models, which prioritizes global coherence over local detail fidelity.

Transformer-based models dominate precision editing but lag in creation tasks. Unified multimodal models employing Transformer architectures (OmniGen, OmniGen2) demonstrate superior performance on precision editing tasks ($\text{Advantage}^{\text{edit}} = +0.91$), particularly replace-identity and add-appearance, where explicit spatial attention to reference regions is critical. Conversely, these models underperform on high-density creation tasks ($k \geq 5$, $\text{Advantage}^{\text{creation}} = -0.68$), possibly because Transformer cross-attention mechanisms scale poorly with the number of conditioning inputs, leading to attention dilution (Wu et al., 2025).

Reference-based models achieve high fidelity on editing but fail to generalize to creation. Reference-based systems (MultiRef, FreeEdit) attain the highest scores on editing tasks (mean 67.8), leveraging spatial correspondence mechanisms such as feature alignment or attention masking to achieve precise concept transfer. However, these models perform poorly on creation tasks (mean 42.1), where no source image provides spatial scaffolding. This limitation underscores a fundamental trade-off: specialized condition-

ing mechanisms optimized for editing workflows sacrifice generality required for *de novo* synthesis.

Design implications. The heterogeneity in model performance across task clusters suggests that no single architecture dominates all aspects of multi-concept generation. Hybrid systems that route tasks to specialized sub-modules—diffusion models for style transfer, Transformers for precision editing, and autoregressive models for creation—may achieve superior aggregate performance by exploiting architectural complementarity. Alternatively, meta-learning approaches that adapt processing strategies dynamically based on task characteristics could mitigate task-specific weaknesses. These directions warrant empirical investigation in future benchmark iterations.

4.5. Ablation Studies and Additional Analyses

Impact of prompt rewriting. To quantify the effect of human prompt rewriting (§3.3), we compare model performance on AI-generated prompts versus human-rewritten prompts for a held-out subset of 300 test cases. Models achieve a mean score of 52.3 on AI-generated prompts, compared to 58.7 on human-rewritten prompts, corresponding to a 12.2% relative improvement. This gain is primarily attributable to reduced ambiguity (instruction-following scores improve from 3.2 to 3.8 on the 5-point scale) and enhanced naturalness (human evaluators rate rewritten prompts as 4.3 versus 3.1 for AI-generated prompts on a 5-point naturalness scale). This finding validates our hybrid pipeline’s design choice to combine AI-assisted scalability with human linguistic refinement.

Sensitivity to reference image quality. We investigate whether concept preservation failures stem from inadequate reference image quality by conducting a controlled experiment: for 200 test cases exhibiting low Material preservation (<0.3), we manually replace low-quality Material reference images (subjective quality <3.5 out of 5) with high-quality alternatives (subjective quality ≥ 4.5) while holding all other factors constant. Material preservation scores improve modestly from 0.26 to 0.34 (30% relative gain), indicating that reference quality contributes to but does not fully explain preservation failures. The persistent difficulty in Material preservation even with high-quality references suggests fundamental representational limitations rather than purely data quality issues.

Cross-model failure correlation. To assess whether failures are systematic or model-specific, we compute pairwise failure overlap: the fraction of test cases on which both models fail (defined as composite score <0.5). The mean pairwise overlap is 61.3%, indicating substantial agreement on difficult cases. Notably, Material omission failures ex-

hibit 73.2% overlap, whereas Identity preservation failures show only 42.7% overlap, suggesting that Material failures are more intrinsic to the task whereas Identity failures reflect model-specific weaknesses. This finding motivates the development of concept-specific evaluation protocols that distinguish intrinsic task difficulty from architectural deficiencies.

5. Conclusion and Future Directions

We have presented MICE-Bench, a comprehensive benchmark designed to rigorously evaluate multi-concept image generation and editing capabilities under heterogeneous constraint compositions. Through a carefully structured four-stage construction pipeline combining AI-assisted planning, human expert rewriting, and concept-directed annotation, MICE-Bench provides 3,769 test cases spanning 17 editing subtasks and 99 multi-concept creation configurations. Our customized evaluation framework, featuring concept-specific verification questions and automated failure attribution, enables fine-grained diagnosis of model limitations that remain obscured in existing single-reference or homogeneous multi-reference benchmarks.

Comprehensive evaluation of eight representative state-of-the-art models reveals systematic performance degradation as constraint density increases, with aggregate scores declining from 68.7 at $k = 2$ to 31.4 at $k = 7$. This non-linear collapse exhibits a pronounced tipping point at the transition from four to five heterogeneous concepts, suggesting fundamental capacity limitations in contemporary architectures. Failure mode analysis identifies concept dominance—wherein Style concepts suppress Material and Appearance in 43.2% of failures—and density-dependent concept omission as the primary bottlenecks. Furthermore, our concept interference matrix quantifies asymmetric suppression effects, demonstrating that Material is the most fragile concept (average suppression 0.27) while Identity remains robust (average suppression 0.09), reflecting systematic imbalances in model prioritization likely attributable to training data distributions and architectural inductive biases.

The diagnostic analyses presented in this work provide actionable insights for future model development. First, the identification of a capacity tipping point at $k \approx 4-5$ motivates architectural innovations such as dynamic capacity allocation via mixture-of-experts mechanisms, hierarchical fusion pipelines that sequentially integrate concepts in order of priority, or explicit reasoning modules that plan compositional strategies before generation. Second, the observed asymmetry in concept interference suggests the utility of concept-specific attention weights or adversarial regularization to enforce balanced preservation across all concept categories. Third, the heterogeneity in task-specific performance patterns—with diffusion models excelling at style

transfer but underperforming on material replacement, and Transformer-based models dominating precision editing but lagging on creation tasks—indicates that hybrid architectures routing tasks to specialized sub-modules may achieve superior aggregate performance.

Beyond model improvements, MICE-Bench establishes a standardized platform for auditing multi-concept generation systems under realistic compositional constraints. The benchmark’s modular design—with separate editing and creation task families, stratified constraint density gradients, and concept-level failure attribution—supports diverse research applications, including identifying training data biases, validating architectural modifications, and benchmarking emerging multimodal foundation models. We release MICE-Bench publicly with evaluation code, annotated test cases, and baseline model outputs to facilitate reproducible research and enable community-driven progress toward robust multi-concept visual synthesis.

Future work may extend MICE-Bench in several directions. First, incorporating additional concept categories such as emotion, lighting, or 3D geometry would broaden coverage of professional creative workflows. Second, expanding to video generation would assess temporal coherence and motion consistency under multi-concept constraints, a critical capability for applications in film production and animation. Third, developing concept-aware training objectives that explicitly encourage balanced preservation during multi-reference generation could mitigate the dominance and omission failures identified in our experiments. Finally, investigating the interplay between constraint density, prompt complexity, and model capacity through theoretical analysis may yield principled scaling laws for multi-concept generation, guiding the design of next-generation architectures capable of reliably composing heterogeneous visual concepts at scale.

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A. Supplementary Material

A.1. Visual Concept Examples

Each of the seven visual concepts in our taxonomy captures a distinct dimension of visual information. Identity encompasses facial geometry including face shape, eye structure, nose contours, and identity-defining features. Style represents global artistic rendering including impressionism, watercolor, oil painting techniques, color harmonies, and brush stroke patterns. Material describes surface properties such as wood grain texture, metallic reflectance, fabric weave patterns, and tactile appearance. Pose specifies body posture, skeletal configuration, limb positioning, and spatial orientation. Scene captures environmental context including indoor/outdoor settings, architectural elements, natural landscapes, and ambient lighting conditions. Object denotes concrete instances with specific geometric and appearance properties such as vehicles, furniture, animals, or architectural structures. Appearance refers to wearable attributes including clothing styles, accessories, makeup, and fashion elements that can be independently modified without altering identity.

A.2. Concept Pool Construction Details

The concept pool construction process involves three interconnected stages: data collection, hierarchical annotation, and quality validation.

Data collection sources. We collect images from multiple curated sources to ensure diversity and quality: (1) **Public datasets** including CelebA and FFHQ for Identity concepts, WikiArt for Style concepts, DTD (Describable Textures Dataset) for Material concepts; (2) **Creative Commons repositories** from Flickr, Unsplash, and Pexels with appropriate licensing; (3) **Specialized collections** for Object and Scene categories ensuring coverage of diverse semantic categories (e.g., vehicles, furniture, indoor/outdoor scenes).

Hierarchical metadata annotation. Beyond concept-level labels, we construct a two-tier hierarchical annotation $\mathcal{M} : \mathcal{C} \rightarrow 2^S$ mapping each concept to fine-grained subcategories:

- **Appearance:** Accessory (hats, glasses, jewelry), Clothes (casual, formal, sportswear), Makeup (natural, dramatic, theatrical)
- **Object:** Animal (mammals, birds, aquatic), Bridge, Building (residential, commercial), Furniture (seating, storage), Plant (flowers, trees), Vehicle (cars, motorcycles), Word (text overlays)
- **Pose:** Background Canny (edge maps), Direction (orientation), Foreground Canny, Lightning (illumination),

Pose (body skeletal)

- **Style:** Style (global artistic), Style Background (environmental rendering), Style Object (object-level stylization)
- **Identity:** Group Photo (multiple persons), Person (individual portraits)
- **Scene:** Bridge, Building, Indoor, Indoor Complex (multi-room), Landscape, Outdoor
- **Material:** Material (surface properties), Texture (patterns and finishes)

This hierarchical structure enables controlled sampling during test case generation, ensuring semantically coherent and physically plausible concept combinations while avoiding incompatible pairings (e.g., "liquid-textured buildings").

Quality validation pipeline. Images undergo a three-stage validation pipeline: (1) **Automated rule-based filtering** removes images below 512×512 resolution, containing visible watermarks, or exhibiting severe compression artifacts; (2) **Automated quality scoring** uses Gemini-3 to evaluate technical quality (sharpness, noise level, color balance) and concept purity (target concept visual dominance), retaining only images scoring $\geq 3.5/5$; (3) **Manual expert review** verifies copyright compliance, content appropriateness (safety filtering), and semantic alignment with assigned concept categories.

From 12,347 initial candidates, 9,011 images passed all validation stages (pass rate: 73.0%). Primary filtering reasons: insufficient resolution (15.2%), quality below threshold (7.3%), concept impurity—multiple concepts present (4.5%). This rigorous pipeline ensures the concept pool provides high-quality, unambiguous reference signals for multi-concept generation evaluation.

A.3. Instruction Following Evaluation Details

Multi-concept prompts frequently contain image references (e.g., "replace the car with the dog in image1"), precluding direct text-based evaluation. We transform such prompts into self-contained textual descriptions via GPT-4V following three transformation principles.

Principle 1: Contextual relevance. Describe only attributes explicitly required by the editing instruction, omitting contextual elements irrelevant to the task. For instance, the instruction "replace the car with the dog in image1" should transform to "replace the car with a golden retriever with brown fur" rather than "replace the car with a golden retriever standing on grass with a blue sky background." This ensures evaluation focuses on task-critical attributes

rather than incidental background details present in reference images.

Principle 2: Attribute transfer for style/material tasks.

For style transfer and material modification tasks, describe artistic characteristics or texture properties rather than subject content. Example transformations:

- Style transfer: “Apply the style from image2” → “Render in impressionist style with visible brush strokes, vibrant color palette, and soft edges”
- Material transfer: “Change material to image3” → “Change surface to polished marble texture with white and gray veining”

This principle ensures evaluation assesses stylistic/material fidelity rather than subject similarity.

Principle 3: Semantic coherence. Generate fluent, grammatically correct descriptions suitable for image generation evaluation. Avoid fragmented lists or technical notation. Example:

- Before: “ID: woman, young, Asian. Hair: black, long. Clothes: red dress”
- After: “A young Asian woman with long black hair wearing a red dress”

Implementation. We provide GPT-4V with the editing instruction I , all reference images $\{r_1, \dots, r_k\}$, and the three transformation principles. GPT-4V performs visual analysis to extract task-relevant attributes and synthesizes coherent textual descriptions D . Gemini-3 then evaluates whether generated image $\hat{y}$ satisfies description D on a 5-point scale: 1 (completely mismatched), 2 (major discrepancies), 3 (partially aligned), 4 (mostly aligned), 5 (perfectly aligned). Final instruction following score: $S_{IF} = \frac{\text{rating}-1}{4}$ normalized to $[0,1]$.

A.4. Automated Test Case Generation Details

Our three-stage automated pipeline progressively refines abstract concept combinations into concrete test cases.

Stage 1: Abstract concept combination planning. For each k -concept combination $\{c_{i_1}, \dots, c_{i_k}\} \subset \mathcal{C}$, GPT-4 generates semantically consistent subcategory combinations based on the hierarchical index $\mathcal{M}$. The planner produces candidate plans:

$$\text{Plan}(\{c_{i_1}, \dots, c_{i_k}\}) = \{(s_{i_1}, \dots, s_{i_k}) \mid s_{i_j} \in \mathcal{M}(c_{i_j}), \text{Feasible}(s_{i_1}, \dots, s_{i_k})\} \quad (1)$$

where $\text{Feasible}(\cdot)$ encodes physical constraints. For instance, Material concepts must be paired with compatible Object or Identity substrates, excluding physically impossible configurations like “metal-textured sky.”

Stage 2: Concrete image sampling and concept-directed captioning.

For each feasible tuple, we sample images and apply concept-directed captioning. For image r_j with concept c_j , GPT-4V generates captions exclusively describing c_j -relevant attributes:

- Identity: facial geometry and skin tone only (omit clothing, background)
- Material: texture and reflectance only (omit object shape)
- Style: artistic properties only (omit subject content)

This ensures downstream evaluation can isolate concept-level failures without conflation.

Stage 3: Visual validation and prompt generation.

GPT-4V performs joint visual-semantic validation, assessing image fusibility via binary judgment. For feasible combinations, it generates fusion logic (e.g., “apply facial features from cap_1 to subject in scene r_3 , rendered in style cap_2 ”), followed by natural-language test prompts. Infeasible combinations (12.7%) are filtered. Output: complete test cases $\langle p, \{r_1, \dots, r_k\}, \{\text{cap}_1, \dots, \text{cap}_k\} \rangle$.

A.5. Diagnostic Analysis Method Formalizations

Concept interference matrix. For concept pair (c_i, c_j) , we compute:

$$M_{ij} = \frac{1}{N_{ij}} \sum_{s \in S_{ij}} (\text{Score}_{c_i}(s, \text{isolated}) - \text{Score}_{c_i}(s, c_j \text{ present})) \quad (2)$$

where S_{ij} contains test cases with both concepts, $N_{ij} = |S_{ij}|$. Positive M_{ij} indicates c_j suppresses c_i ; negative values indicate facilitation. This reveals “dominant concepts” (high row sums) and “fragile concepts” (high column sums).

Constraint density tipping point detection. We compute marginal impact and acceleration:

$$\begin{aligned} \Delta_C(k) &= \text{Score}(k) - \text{Score}(k-1) \\ \Delta^2(k) &= \Delta_C(k) - \Delta_C(k-1) \end{aligned}$$

Large negative Δ^2 values identify accelerating degradation regions. For concept-level analysis, we decompose transitions by specific added concepts to identify “bottleneck triggers.”

Task-specific capability profiling. Construct $P \in \mathbb{R}^{8 \times 17}$ performance matrix. Apply k -means ($k = 3$) to partition tasks by difficulty. Compute relative advantage:

$$\text{Advantage}_i^{\text{class}} = \frac{\text{Score}_i^{\text{class}} - \overline{\text{Score}}^{\text{class}}}{\sigma^{\text{class}}}$$

Positive values indicate architectural strengths; negative values reveal weaknesses.

Table 3 presents the detailed distribution of the MICE-Bench concept pool. The concept pool comprises 9,011 high-quality images spanning seven concept categories. Each category is further subdivided into 2–7 fine-grained sub-categories to ensure diversity and comprehensive coverage. This hierarchical organization enables semantically coherent concept sampling during test case generation (§3.3), thereby supporting controlled combinatorial exploration of heterogeneous concept compositions.

The distribution reflects the relative frequency of different concept types in professional creative workflows, with Appearance and Object concepts comprising the majority of the pool due to their prevalence in real-world editing scenarios. Material concepts, while less numerous, are selected to represent diverse texture properties (wood, metal, fabric, stone) critical for evaluating fine-grained attribute transfer.

A.6. Quality Control Standards

Table 4 enumerates the quality control standards applied during concept pool construction. All images undergo dual-stage validation combining rule-based filtering and AI-assisted assessment to ensure high visual quality, concept purity, and ethical compliance.

Pass rate statistics. From an initial collection of 12,347 candidate images, 9,011 images passed quality control, corresponding to a pass rate of 73.0%. The primary filtering reasons were: insufficient resolution (15.2%), quality below threshold (7.3%), and concept impurity (4.5%). This rigorous filtering ensures that the concept pool provides reliable reference signals for multi-concept generation evaluation.

A.7. Human Prompt Rewriting Principles

Table 5 describes the four core principles guiding human prompt rewriting. Six professional image editors (mean experience: 6.2 years) manually rewrite all AI-generated prompts to ensure linguistic naturalness and task feasibility while preserving semantic validity.

Quality assurance. Each prompt is assigned to at least two independent annotators, with inter-annotator agreement measured via Fleiss’ Kappa = 0.79, indicating substantial agreement. If a majority of annotators (≥ 3 out of 6) judge a

combination infeasible, the test case is removed (removal rate: 12.7%). This dual-layer quality control ensures that MICE-Bench prompts reflect authentic user instructions while maintaining semantic validity.

A.8. Concept-Directed Captioning Examples

Table 6 presents examples of concept-directed captioning for each concept category. This specialized captioning strategy ensures that each reference image’s description contains only the target concept information, thereby supporting precise failure attribution during evaluation (§3.4).

Contrast with generic captioning. Generic VLM captioning would produce descriptions such as “An Asian woman in a dark blue suit standing indoors, smiling, with a white wall background,” which conflates Identity + Appearance + Pose + Scene information. Such conflation undermines precise failure attribution, as evaluation cannot isolate concept-level preservation failures. Concept-directed captioning constrains the annotation scope to ensure evaluation independence across concepts.

A.9. Concept Preservation Verification Question Validation

All automatically generated concept preservation verification questions (§3.4) undergo validation by three computer vision experts with an average of 4.5 years of experience in multimodal generation research. Experts independently audit all questions to ensure they meet four validation criteria:

- **Contrastiveness:** Questions explicitly require comparing the generated image with the reference image, rather than merely verifying presence.
- **Attribute Accuracy:** Attribute descriptions are accurate for the concept category (Identity focuses on facial features, Style focuses on artistic style).
- **Unambiguity:** Question phrasing is clear with no multiple interpretations.
- **Answerability:** VLMs can provide definitive Yes/No answers based on visual information.

Validation results. Of the 3,769 automatically generated questions, 3,442 passed validation (pass rate: 91.3%). Failed questions were primarily due to: overly broad phrasing (4.2%), inaccurate attribute description (2.8%), and ambiguous phrasing (1.7%). Failed questions were manually revised by experts before inclusion in the evaluation protocol. This rigorous validation ensures that concept preservation scores accurately reflect model capabilities rather than evaluation artifacts.

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Table 3. Distribution of the MICE-Bench concept pool across seven concept categories and 28 subcategories.

Concept	Subcats	Images	Subcategory Names
Appearance	3	2,634	Accessory, Clothes, Makeup
Object	7	3,053	Animal, Bridge, Building, Furniture, Plant, Vehicle, Word
Pose	5	1,578	Background Canny, Direction, Foreground Canny, Lightning, Pose
Style	3	644	Style, Style Background, Style Object
Identity	2	599	Group Photo, Person
Scene	6	338	Bridge, Building, Indoor, Indoor Complex, Landscape, Outdoor
Material	2	165	Material, Texture
Total	28	9,011	—

Table 4. Quality control standards for concept pool construction.

Standard Category	Specific Requirements	Validation Method
Resolution	$\geq 512 \times 512$ pixels	Automated rule-based filtering
Image Quality	No visible noise, artifacts, or blur	Gemini-3 automated scoring ($\geq 3.5/5$)
Concept Clarity	Target concept occupies dominant position in the image	Gemini-3 semantic detection
Copyright Compliance	Sourced from public datasets or Creative Commons licenses	Manual review + source dataset verification
Content Appropriateness	No sensitive, violent, or adult content	Gemini-3 safety filter
Concept Purity	Single-concept constraint (image contains only one dominant concept)	GPT-4V concept-directed annotation verification

A.10. Additional Experimental Details

A.10.1. MODEL HYPERPARAMETERS

Table 7 lists the inference hyperparameters used for each evaluated model. All settings follow official recommendations to ensure fair comparison and ecological validity.

A.10.2. EVALUATION PROMPT TEMPLATES

For concept preservation verification (§3.4), we provide example question templates for each concept category:

- **Identity:** “Does the person in the generated image have the same facial features (face shape, eyes, nose, mouth) as the person in [reference image]?”
- **Style:** “Does the generated image adopt the same artistic style (brush strokes, rendering technique, color palette) as [reference image]?”
- **Material:** “Does the object in the generated image have the same material texture (surface properties, reflectance) as the object in [reference image]?”
- **Pose:** “Does the subject in the generated image have the same pose (body posture, skeletal configuration) as the subject in [reference image]?”
- **Scene:** “Does the generated image have the same background scene (environment, spatial layout, lighting) as [reference image]?”

- **Object:** “Does the object in the generated image have the same appearance (shape, color, geometric properties) as the object in [reference image]?”

- **Appearance:** “Does the person in the generated image wear the same clothing and accessories as the person in [reference image]?”

These templates are instantiated with task-specific details extracted from the editing instruction via GPT-4 parsing.

A.11. Broader Impact and Ethical Considerations

Intended use. MICE-Bench is designed as a research benchmark to evaluate and improve multi-concept image generation and editing models. The benchmark supports diagnostic analysis of model limitations, identification of failure modes, and development of more robust generation systems for professional creative workflows.

Potential misuse. As with all generative modeling benchmarks, MICE-Bench could potentially be misused to develop systems for creating deceptive or harmful content. We emphasize that the benchmark is intended solely for research purposes to advance the understanding of multi-concept generation. All images in the concept pool are sourced from public datasets or Creative Commons-licensed repositories with appropriate attribution.

Table 5. Principles for human prompt rewriting and quality assurance.

Principle	Specific Requirements	Example
Semantic Preservation	Maintain the core intent and concept composition of the original AI-generated prompt without altering task essence	AI: “A person wearing a hat” → Human: “A person in a red coat wearing a brown hat”
Ambiguity Elimination	Remove expressions admitting multiple interpretations to ensure instruction uniqueness	AI: “Style similar to image2” → Human: “Apply the watercolor painting style from image2”
Naturalness	Use colloquial, user-typical language rather than technical or machine-generated phrasing	AI: “Apply identity from image1 to subject” → Human: “Replace the subject’s face with the person in image1”
Feasibility Assessment	Expert judgment of physical plausibility; remove infeasible combinations	Infeasible: “Change the cat’s material to liquid metal” → Remove test case

Table 6. Examples of concept-directed captioning across seven concept categories.

Concept Type	Annotation Constraint	Annotation Example
Identity	Describe only facial geometry, skin tone, hairstyle; ignore clothing and background	“An Asian woman with a round face, long straight black hair, brown eyes, and a smiling expression”
Style	Describe only artistic style, brush strokes, color tone; ignore specific subject matter	“Watercolor painting style with soft brush strokes, light blue and pink cool tones, high saturation”
Material	Describe only texture, color, material type; ignore object shape	“Dark brown wood texture with rough surface, visible wood grain, matte finish”
Pose	Describe only posture, skeletal structure, spatial relations; ignore identity and clothing	“Person standing upright with arms crossed over chest, head slightly tilted to the right”
Scene	Describe only environmental background and scene context; ignore foreground subjects	“Indoor environment with white walls, wooden floor, natural light through window, minimalist style”
Object	Describe only object instance features; ignore background and material details	“Red sedan with four-door design, streamlined body, black tires”
Appearance	Describe only clothing, accessories, and wearable attributes; ignore identity and pose	“Dark blue suit jacket, white shirt, black tie, brown leather shoes”

Bias and fairness. The concept pool construction process includes diversity considerations across identity attributes (age, gender, ethnicity) for Identity concepts, and representational balance across material types, object categories, and artistic styles. However, as with all large-scale datasets, residual biases may remain. We encourage future work to extend MICE-Bench with additional diversity metrics and fairness-aware evaluation protocols.

Privacy considerations. All Identity concept images are sourced from publicly available face datasets (e.g., CelebA, FFHQ) that have been widely used in prior computer vision research. No private or sensitive data is included in the concept pool. Nevertheless, we acknowledge ongoing discussions regarding the ethical use of face datasets and commit to updating the benchmark in alignment with evolving community standards.

Table 7. Inference hyperparameters for evaluated models.

Model	Steps	Guidance	Temp.	Resolution
FLUX.2	50	7.5	—	1024×1024
Stable Diffusion XL	50	7.5	—	1024×1024
OmniGen	50	7.5	—	1024×1024
OmniGen2	50	7.5	—	1024×1024
Qwen-Edit	—	—	1.0	1024×1024
Step1X-Edit	—	—	1.0	1024×1024
MultiRef	50	7.5	—	1024×1024
FreeEdit	50	7.5	—	1024×1024